

AADHITHYA R

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PROFESSIONAL SUMMARY

AI Engineer focused on transforming complex business problems into scalable AI-driven solutions. Passionate about analyzing operational workflows, uncovering inefficiencies, and building intelligent systems that automate repetitive tasks, streamline decision-making, and improve customer experiences. Experienced in developing production-grade LLM applications, machine learning pipelines, intelligent automation systems, and cloud-based AI solutions with a strong emphasis on measurable business impact and long-term maintainability.

TECHNICAL SKILLS

AI/ML & LLMs: Machine Learning, LLMs, RAG, AI Agents, NLP, Prompt Engineering, LangChain, MCP, Tesseract OCR, AWS Textract

Fine-Tuning & Research: HuggingFace, PEFT, TRL, QLoRA, LoRA, Mistral 7B

Systems Design: REST API Design, Microservices, Request Queuing, Celery, Concurrency Management

Open Source Tools: Flowise, Langflow, Langfuse, Chatwoot, Zammad, Typebot, Directus, Ollama

Languages & Frameworks: Python, FastAPI, Streamlit, REST APIs, OpenAI SDK, Anthropic SDK, LaTeX

Databases & Storage: PostgreSQL, MongoDB, FAISS, PgVector, Supabase, SQLite

Cloud & DevOps: AWS (S3, Bedrock, SageMaker), Docker, RunPod, Git

EXPERIENCE

SPAN TECHNOLOGY SERVICES PRIVATE LTD.

January 2024 – Present

Junior AI Engineer

Erode, India

• Architected MCP-Powered Multi-Tool Assistant with Context-Aware Escalation:

- New users struggled to navigate the tax-filing platform even with guided in-app navigation; built an MCP-powered chatbot enabling direct natural-language queries with intent-based tool routing across 10+ internal tax application APIs
- Enriched every request with real-time user and page context via API for personalized, context-aware responses
- Designed auto-escalation to live agents with full conversation history on unresolved queries – improving resolution rate by 30%

• Anything to Return – Automated Multi-Format Return Generation:

- Clients received tax data across multiple disparate formats – PDF, Excel, CSV, Image, Parquet, ORC, JSON, and FIRE – requiring manual filing and consolidation before a return could be prepared
- Led a team of 2 to design and build a system that automatically extracts data from any supported format and generates a structured return, using Celery-based task queuing to handle concurrent processing requests
- Built a validation step allowing users to review the generated return before transmission, ensuring accuracy while eliminating manual data entry

• Engineered Multi-Engine Document Extraction Pipeline – 80% Manual Effort Reduction:

- Tax forms (W9, W8, W8-BEN, 1095-C, 1042) arrived primarily as image-based PDFs requiring manual data entry
- Built independently tuned extraction endpoints per form type, selecting Tesseract OCR or AWS Textract based on layout complexity
- Implemented Celery-based request queuing to prevent single-user resource monopolization during concurrent multi-file uploads, reducing manual data entry by 80%

• Shipped Production Tax-Filing Chatbot with Auto-Retraining Pipeline – 65% Support Reduction:

- Frequent tax regulation updates required manual re-embedding of content into the vector database – time-consuming and error-prone at scale
- Built a Directus-based auto-retraining pipeline detecting content changes and selectively re-embedding only modified chunks, gated through Slack approval
- Deployed LLM-powered chatbot via Flowise handling 1,000+ daily queries, reducing support tickets by 65%

• Deployed Self-Hosted AI Infrastructure – Eliminated Recurring SaaS Costs:

- Reliance on third-party SaaS tools for chat, support, and automation drove recurring subscription costs at scale
- Self-hosted Flowise, Langfuse, Zammad, Chatwoot, and Typebot via Docker, eliminating third-party SaaS dependency across internal and production environments
- Developed 10+ modular APIs for conversation summarization, sentiment classification, ticket management, and data validation

PROJECTS

Multi-Format Automated Tax Return Generator:

- Users receiving tax data across inconsistent formats (PDF, Excel, CSV, Image, Parquet, ORC, JSON, FIRE) had to manually consolidate and prepare each return
- Built an automated pipeline that ingests any supported format, extracts and validates the data, and generates a transmit-ready return post-payment
- Validated extraction accuracy and pipeline reliability across sample datasets spanning all 8 supported formats

Job Hunt MCP Server & Multi-LLM Client:

- Built an MCP server that automates the end-to-end job search pipeline – searches LinkedIn and Indeed simultaneously, scores listings against a resume using a multi-signal ranking system (ATS score, keyword match, hire probability, seniority and salary fit)
- Exposed 10 MCP tools covering cover letter generation, resume improvement suggestions, application tracking, and side-by-side job comparison – all triggered via natural language
- Built a companion chat client supporting Anthropic SDK, OpenAI SDK, Gemini, and Ollama – client spawns the MCP server as a subprocess, connects via stdio, and routes tool calls automatically across any provider

LLM Fine-Tuning with QLoRA on Mistral 7B:

- Fine-tuned Mistral-7B-Instruct using QLoRA (4-bit NF4 quantization + LoRA adapters) on the Alpaca dataset for instruction-following behaviour
- Implemented the full fine-tuning pipeline from scratch – quantization config, LoRA adapter attachment, dataset formatting, and supervised fine-tuning using HuggingFace PEFT and TRL
- Trained only 0.047% of total parameters (3.4M of 7.2B) via LoRA, reducing memory footprint by 8x compared to full fine-tuning while achieving stable loss convergence from 1.33 to 0.85

Synthetic Customer Data Generator using PostgreSQL and Faker:

- Created a Python utility to simulate and insert realistic customer order data into PostgreSQL databases using Faker for testing and development
- Structured scalable database schemas and auto-populated 100+ diverse records, reducing testing data preparation time by 80%
- Ensured secure and asynchronous database connectivity across environments by leveraging asyncpg and environment-based configuration

CERTIFICATIONS

AWS Certified AI Practitioner	Amazon Web Services	2024
Machine Learning Foundations	AWS Educate	2024
Build Your Own Chatbot	DeepLearning.AI	2024
Foundation in Programming and Data Science	IIT Madras	2023

EDUCATION

Bachelor of Computer Science & Engineering, Easwari Engineering College	CGPA - 9.5 (2020 – 2024)
Gold Medalist – Department of Computer Science & Engineering	

PUBLICATIONS

Docker Build Not Updating? The Build Cache is Probably Why.	Medium
Tired of Re-Embedding Manually? Here’s How We Fixed It for Good	Medium